

JKBOSE 2026

CLASS 11TH

Bhaichara
ON
TOP.

#XUV ♥

HYDROCARBONS

LECTURE-07 CHEMISTRY

VISHAL SIR

JKBose Wallah



Topics To Covered

- 1** **Chemical Properties of Alkenes**
- 2** **Structure of Triple Bond Alkenes**
- 3** **Methods of Preparation of Alkynes**
- 4** **Phy. of Chem. Properties of Alkynes**

CHEMICAL RXNS OF ALKENES.

↓
EAR

Addition of X_2

" H_2

" HX

" H_2SO_4

" H_2O

★

✓

Oxidation

Most Imp

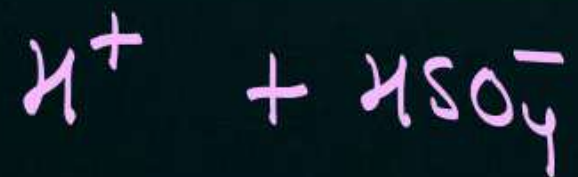
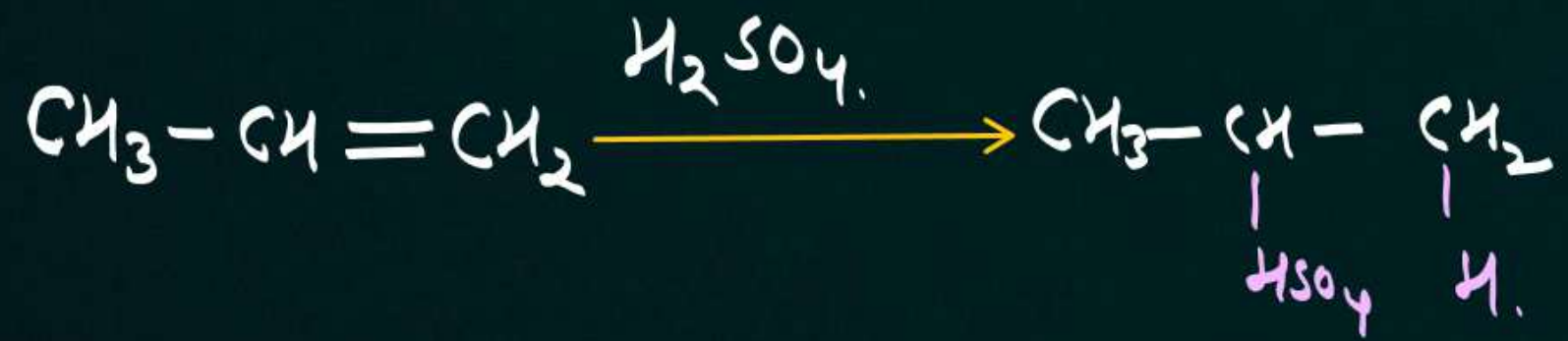
OZONOLYSIS.

POLYMERISATION.

1) ELECTROPHILIC ADDITION RXN.

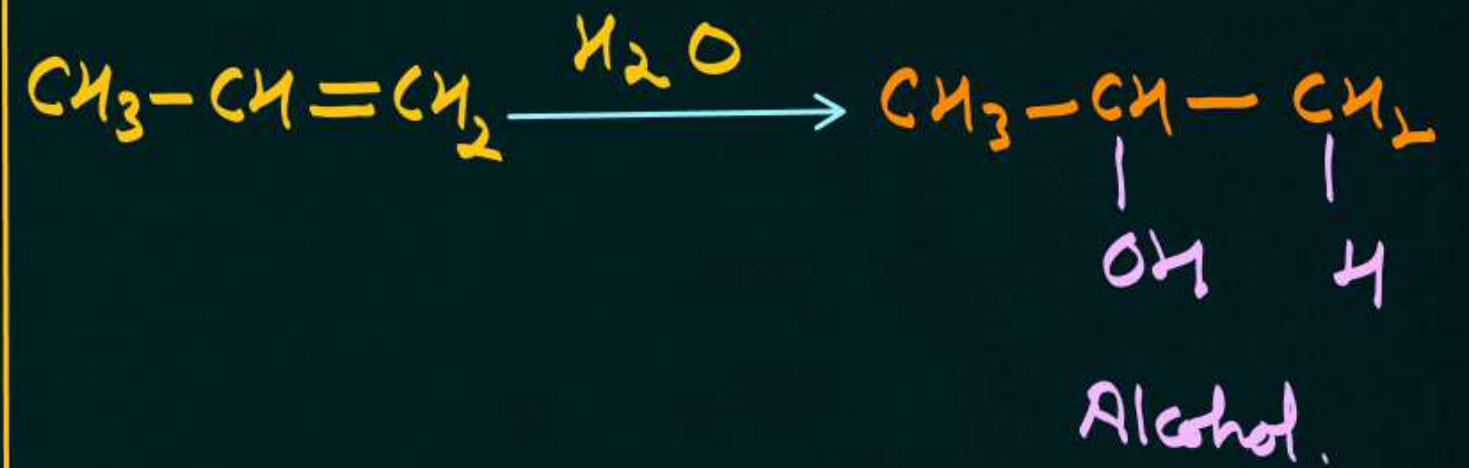
d) Addition of H_2SO_4 .

Addition of H_2SO_4 to alkenes takes place according to Markovnikov Rule



e) Addition of H_2O .

It also takes place according to Markovnikov rule & alcohol is formed.

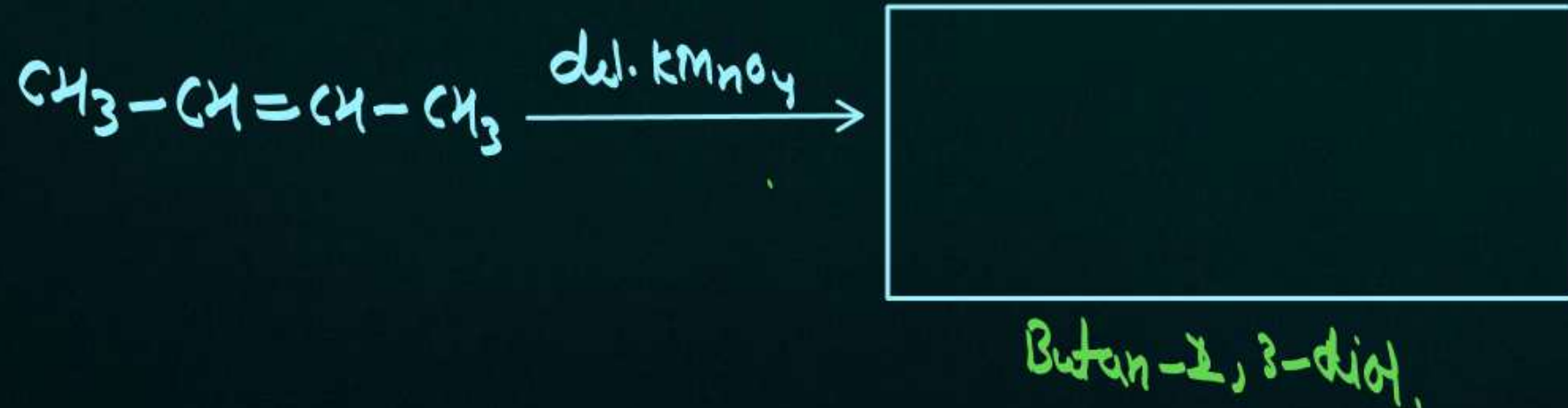


Chemical Properties of Alkenes

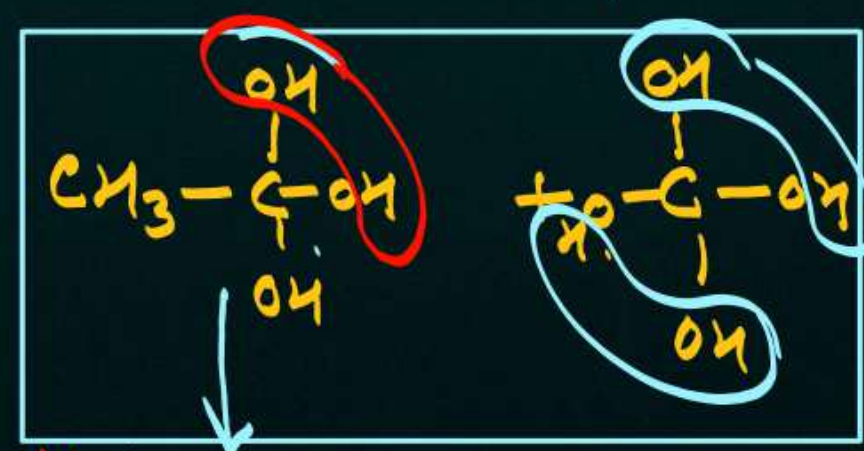
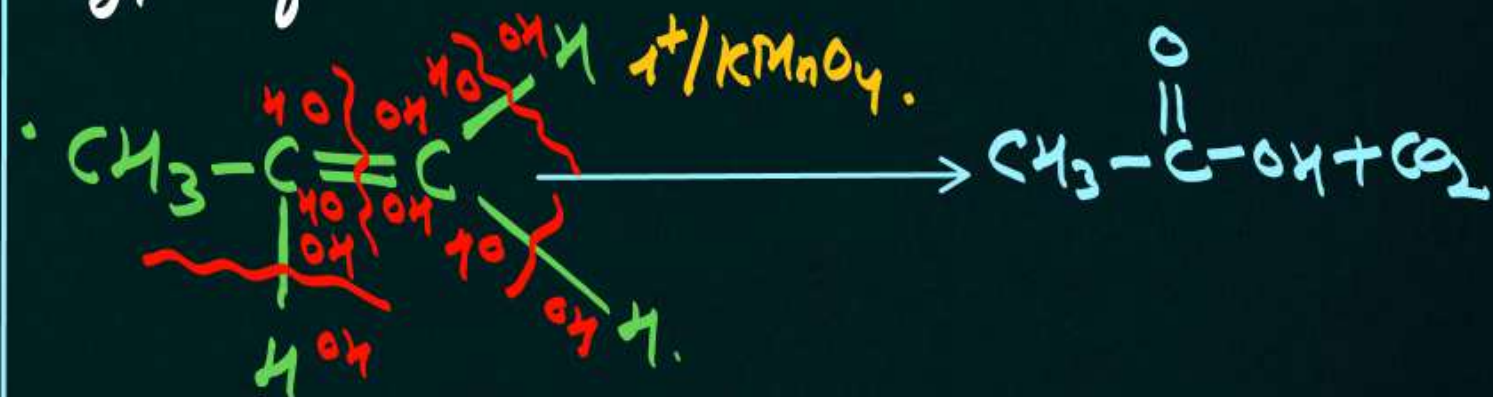


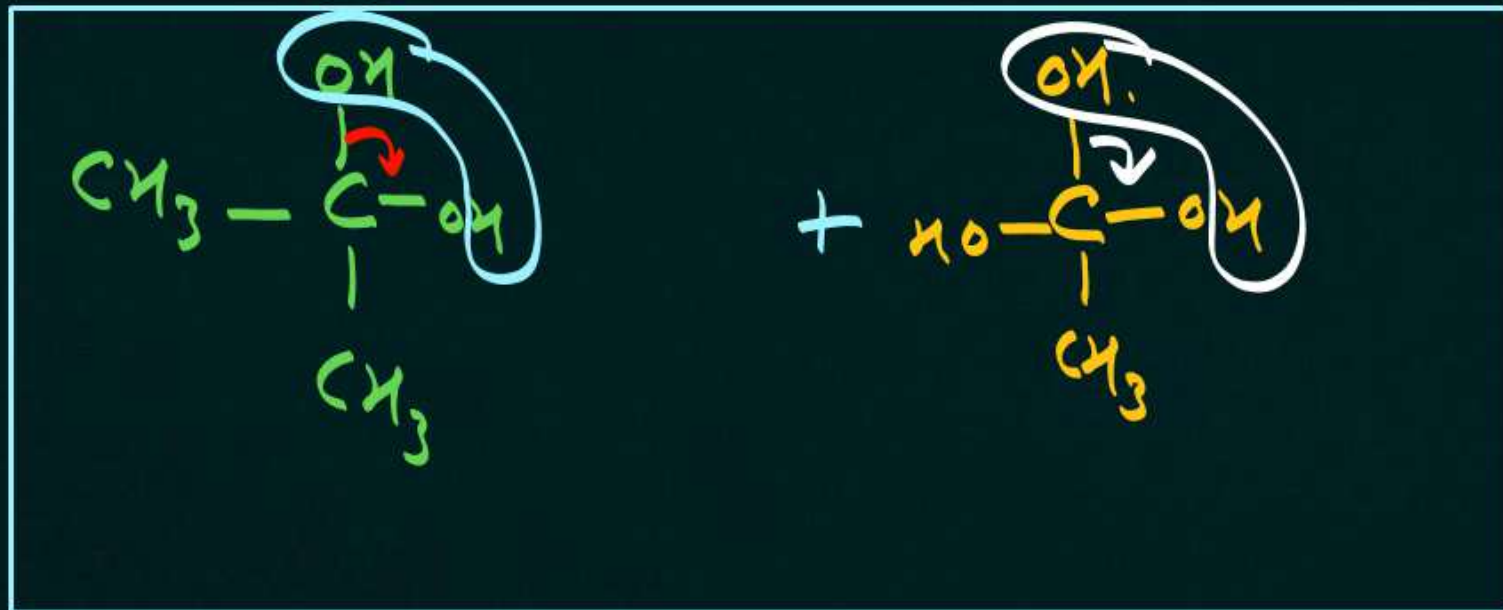
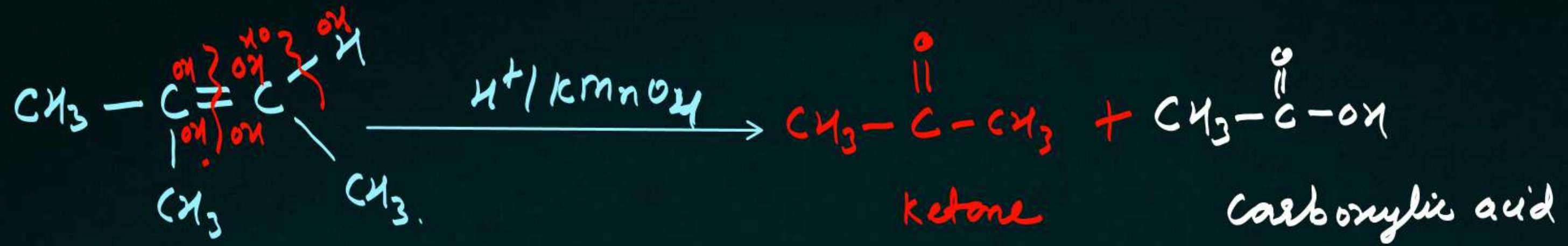
2. Oxidation

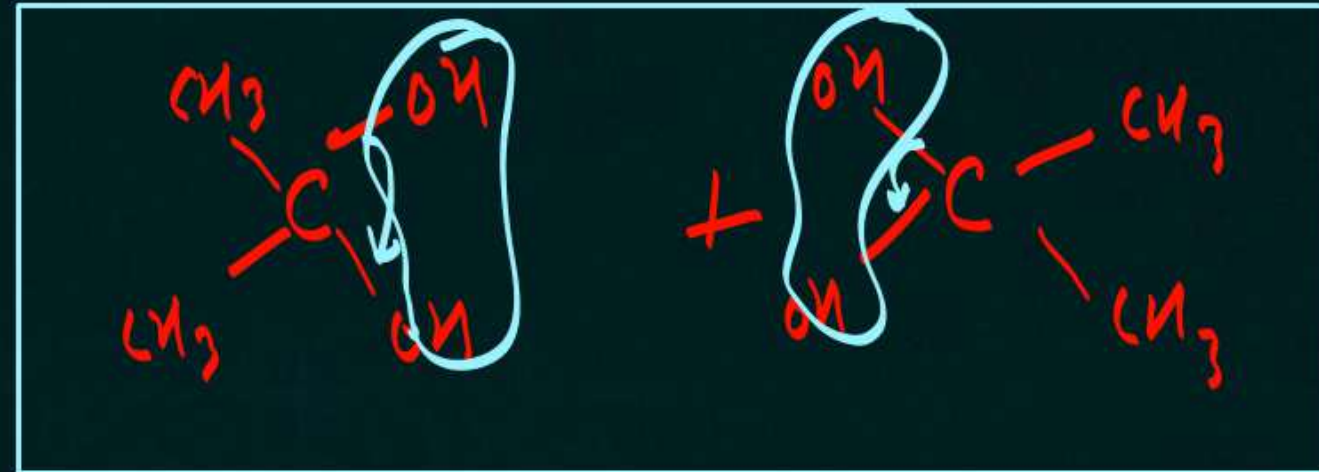
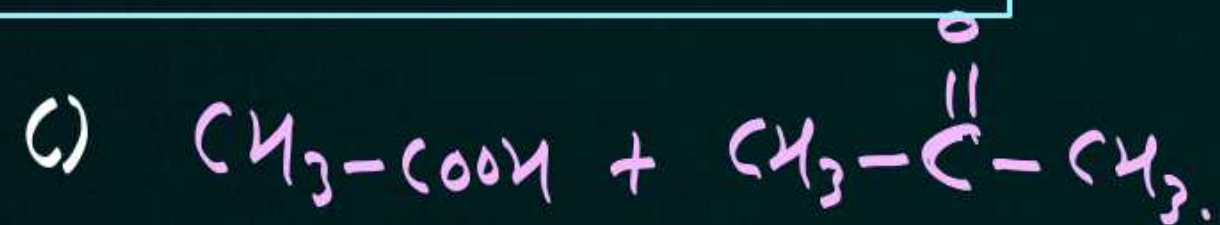
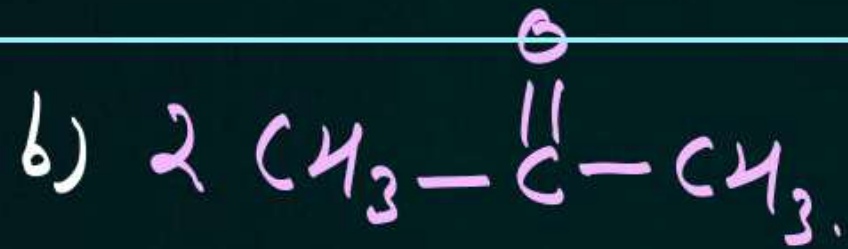
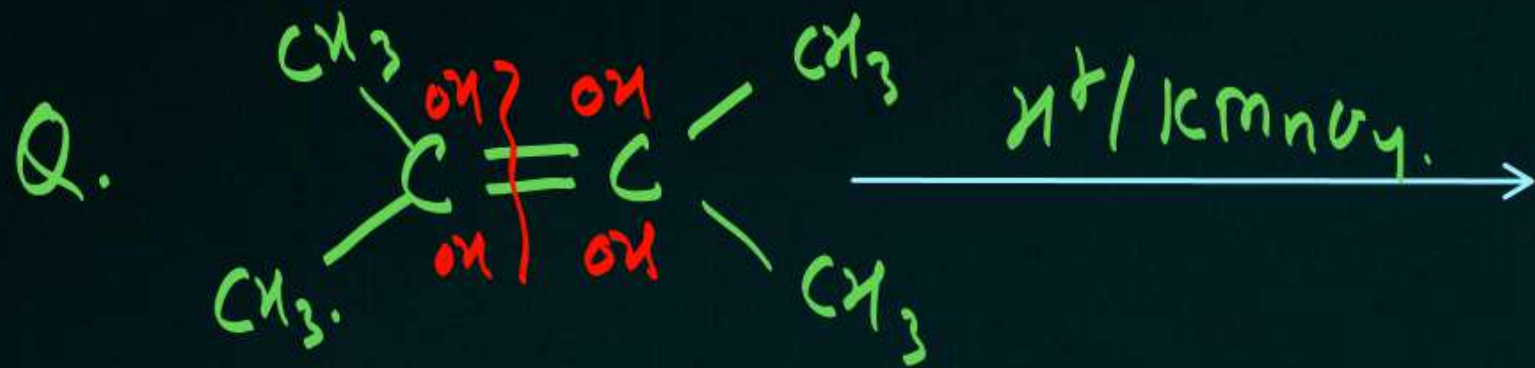
a) Oxidation of alkenes with the help of oxidising agent dil. KMnO_4 (Weak) produces vicinal diols.



b) Oxidation of alkenes with strong oxidising agents like acidified KMnO_4 (Potassium permanganate) or $\text{K}_2\text{Cr}_2\text{O}_7$ (Potassium dichromate) produces different products depending upon the type of alkene.



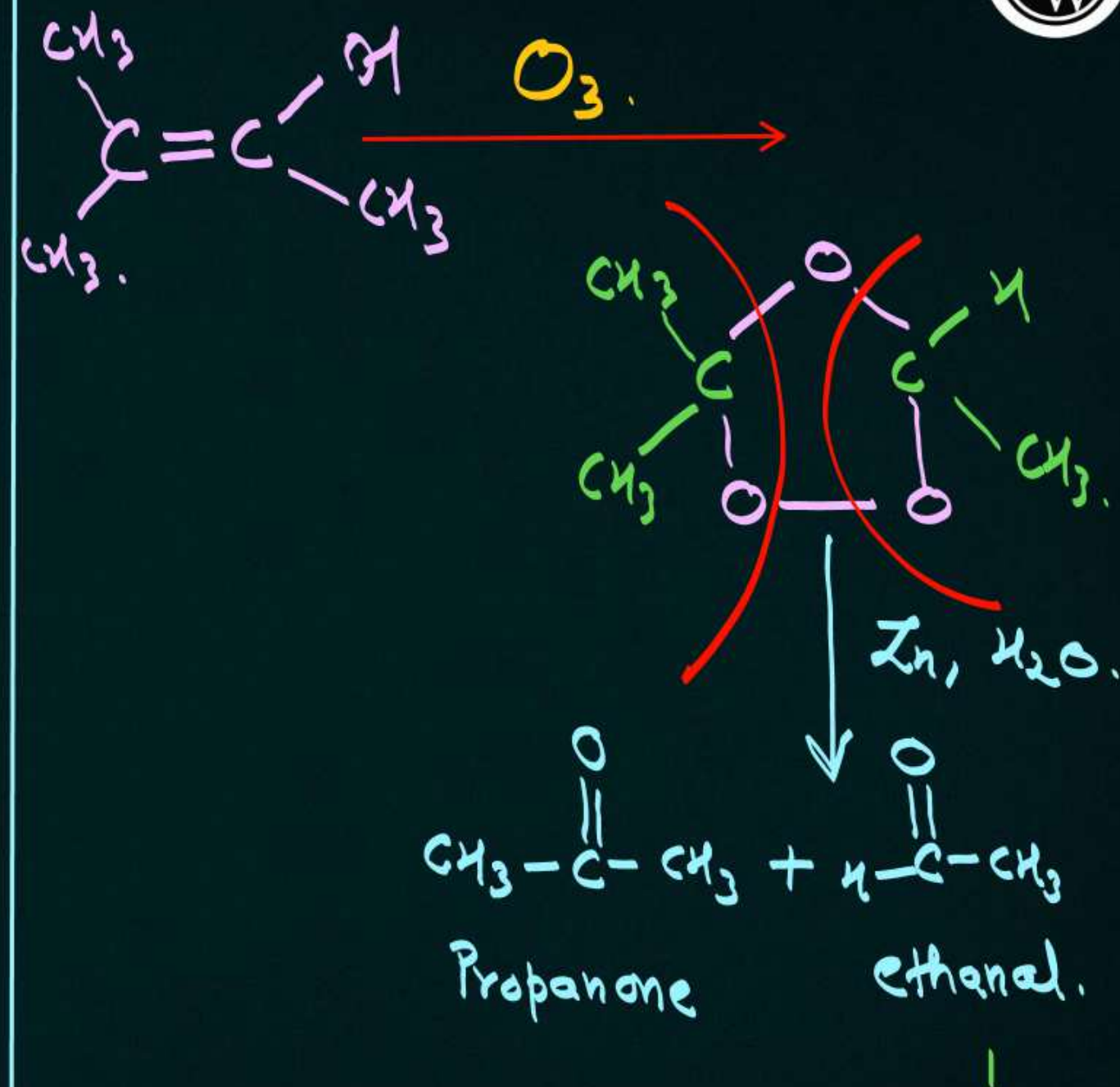
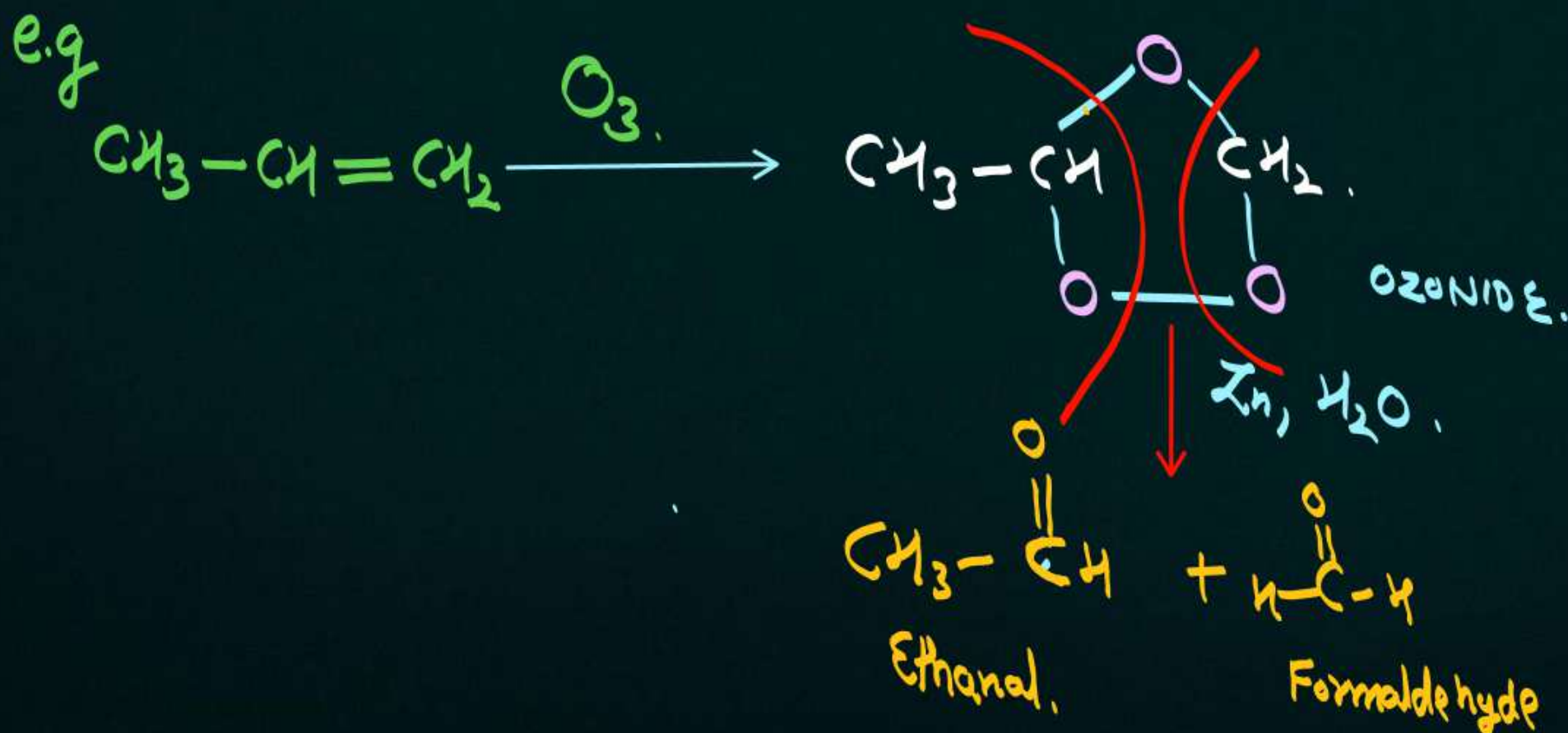


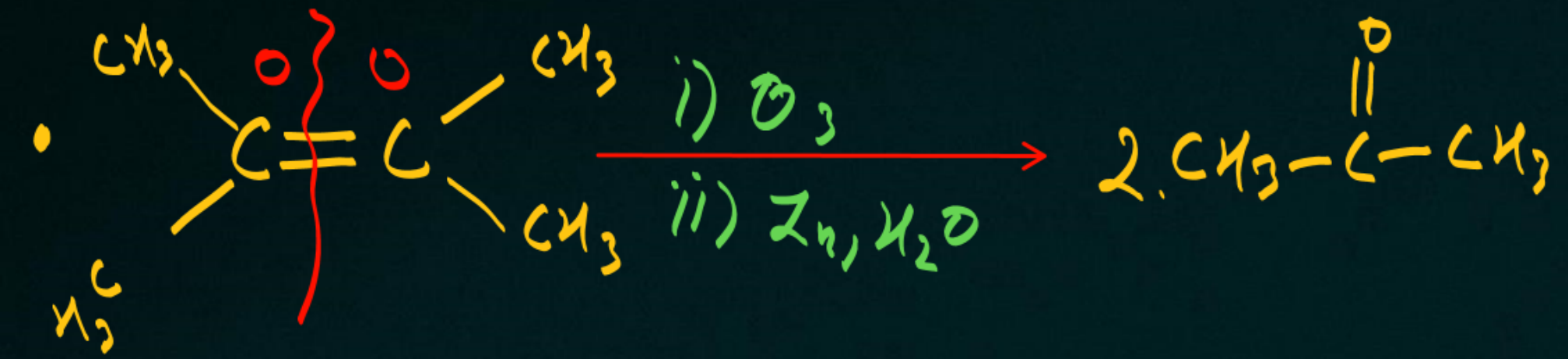


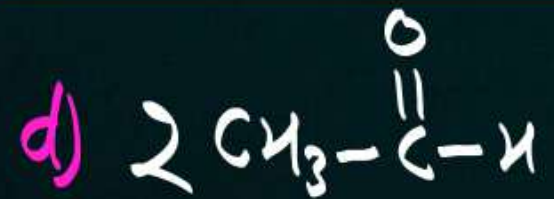
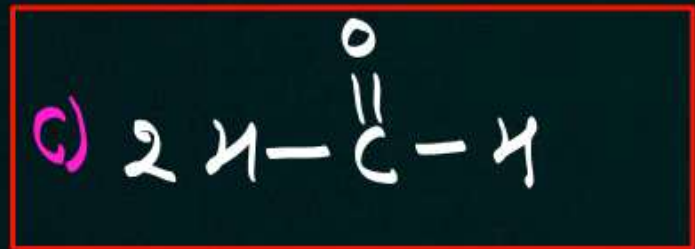
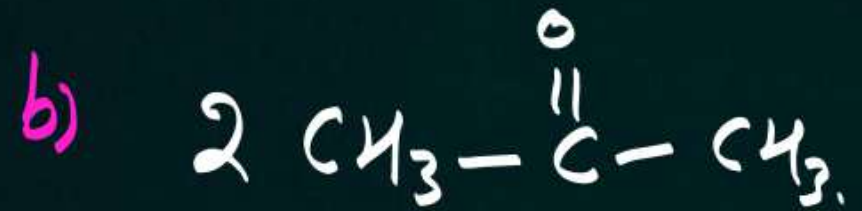
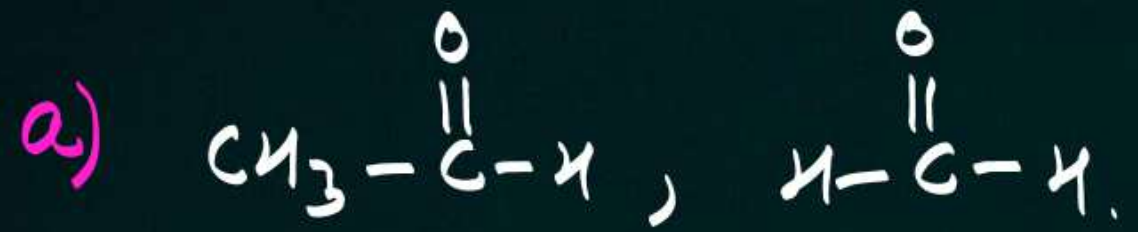
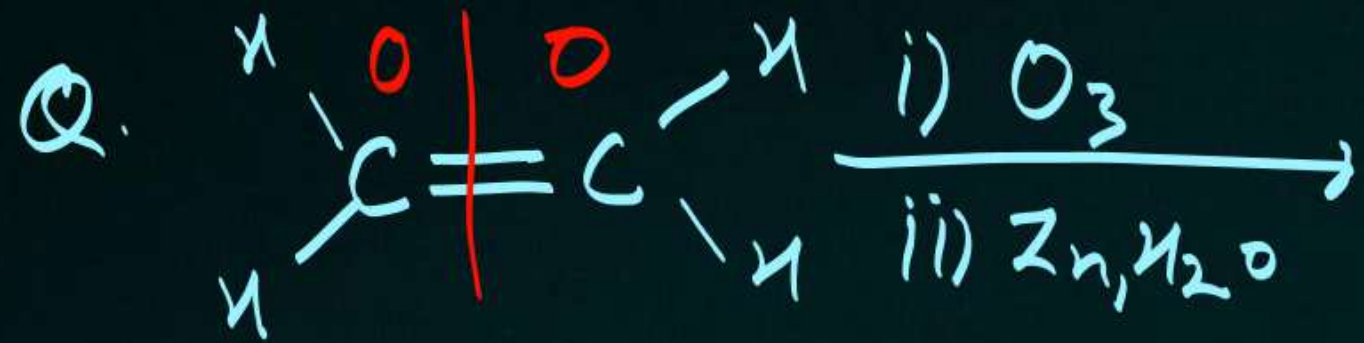
Chemical Properties of Alkenes

3. Ozonolysis

- Alkenes react with O_3 to give ozonide which on reduction with $Zn-H_2O$ cleaved to give smaller molecule, e.g.



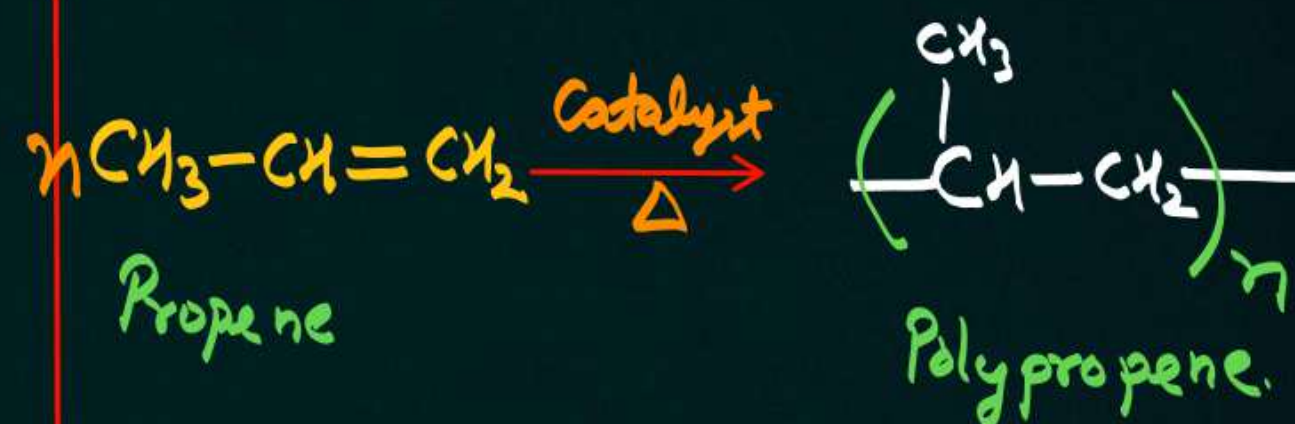
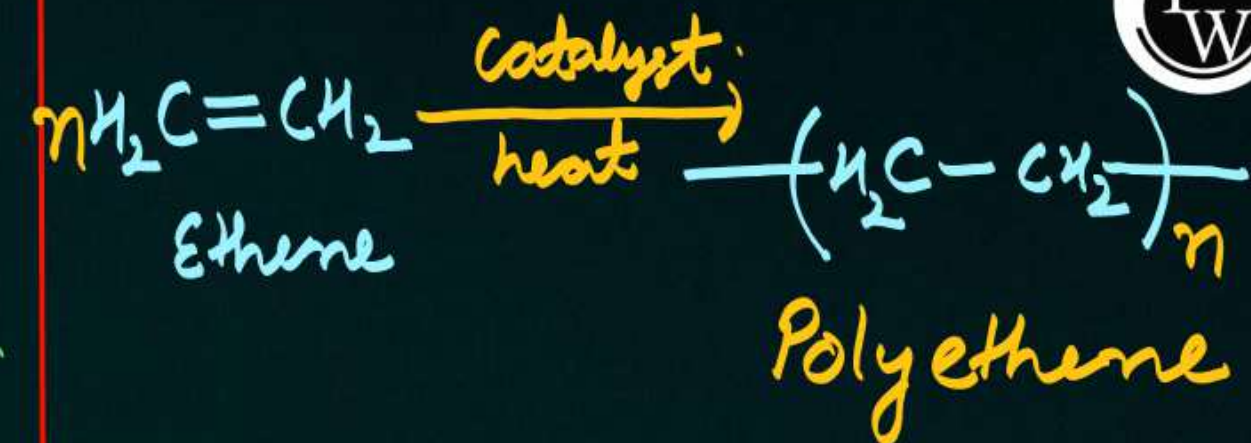




Chemical Properties of Alkenes

4. Polymerisation

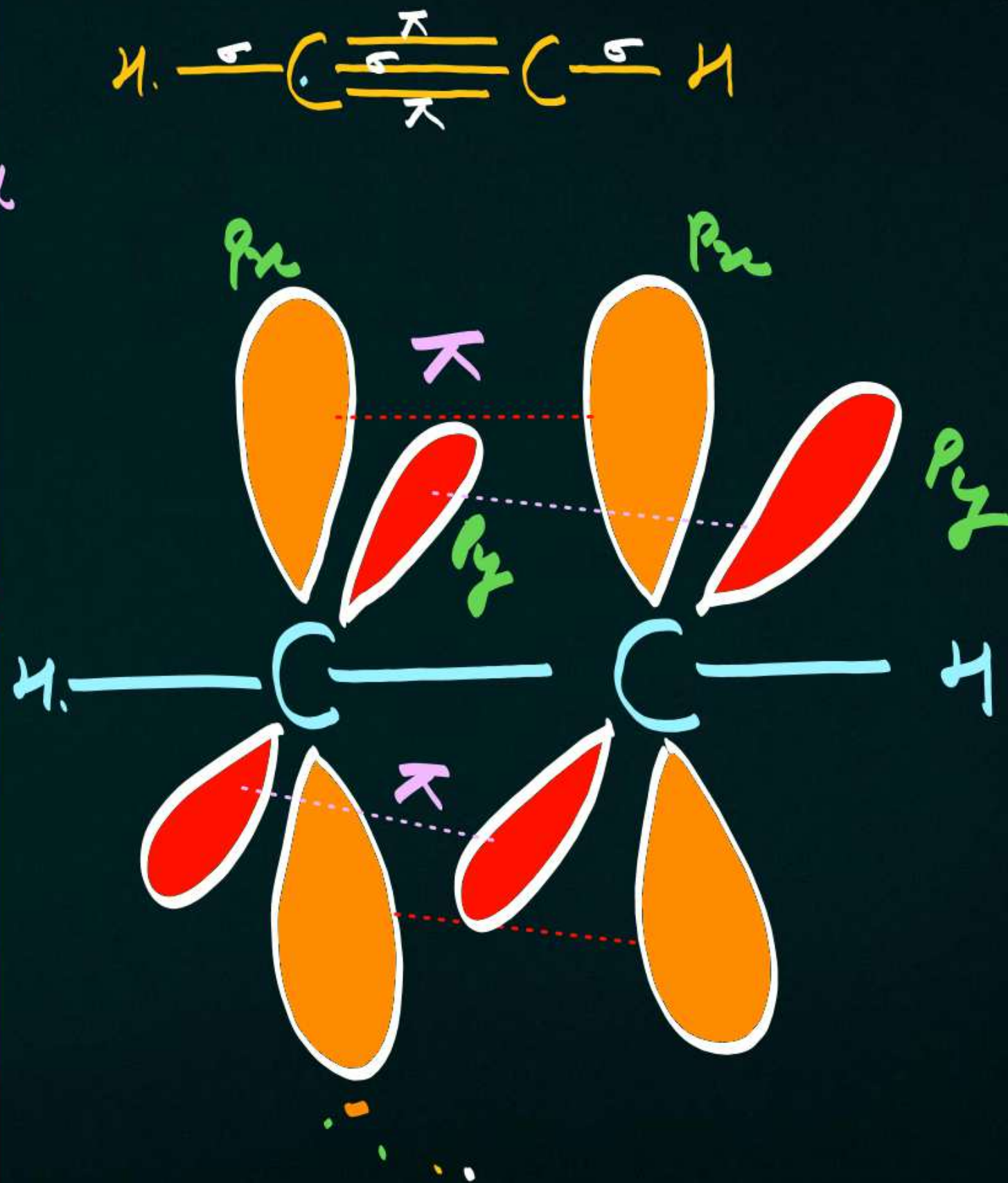
- When a large number of **monomers** (small molecules) combine to give a molecule bearing high molecular mass, called the polymer, the process is called **polymerisation**.
- The polymerisation reactions can be seen as



Alkynes

Structure of Alkyne

- Let us take the example of ethyne.
- Each C has 2 σ & 2 π bonds i.e. sp hybridised then it is linear. (Cylindrical)
- Each C has 2 σ bonds, $\text{C}-\text{C}$ sigma is formed by the overlap of sp hybrid orbitals of carbon & $\text{C}-\text{H}$ sigma bond is formed by the overlap of sp hybrid orbital of C & s -orbital of H .
 - Two $\text{C}-\text{C}$ π bond are formed by side wise overlap p_x-p_x & p_y-p_y orbitals of adjacent carbon.



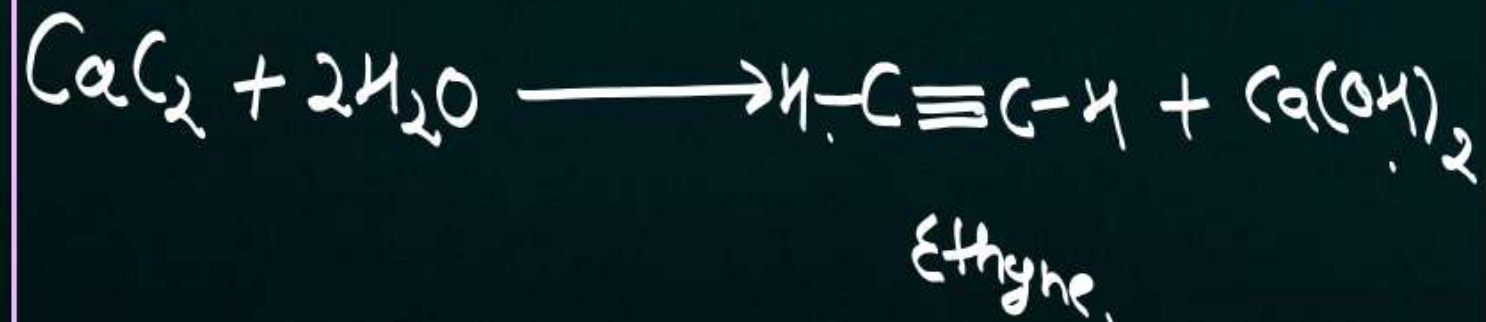
Methods of Preparation of Alkynes

(a) From Calcium Carbide

CaC_2 is formed by heating CaCO_3 which produces CaO & CaO on reaction with 'C' produces Calcium Carbide.

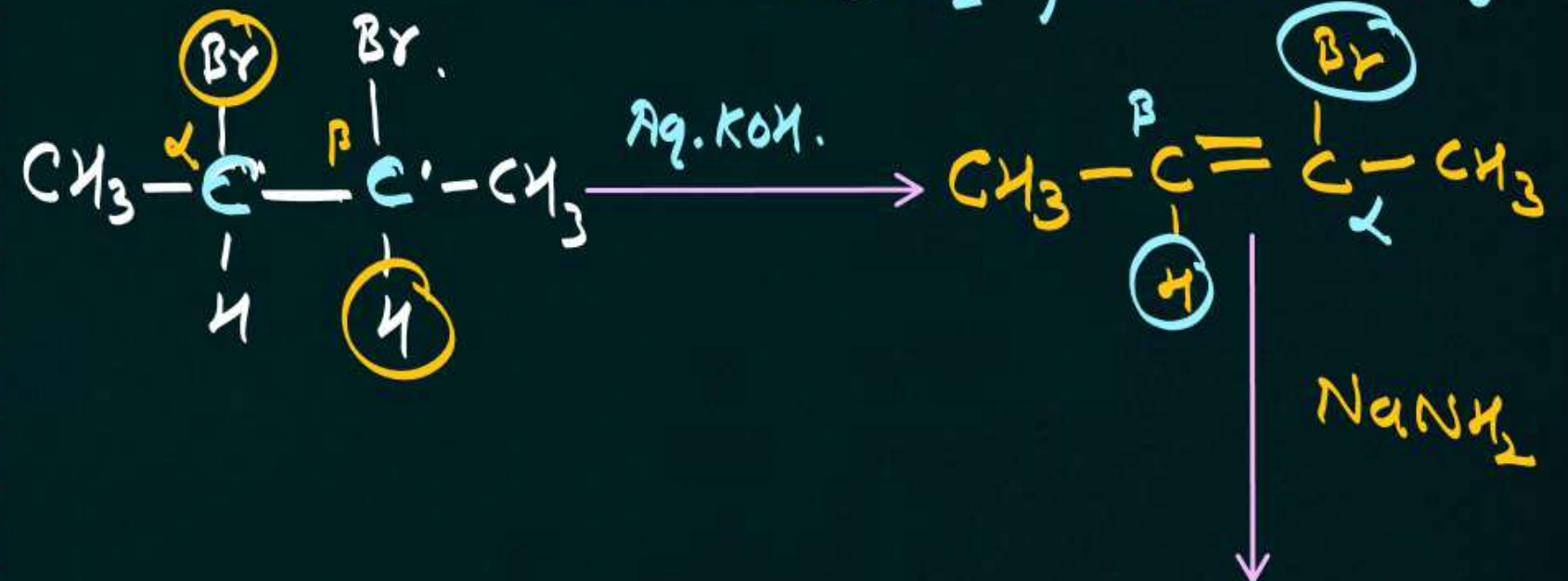


Calcium on reaction with H_2O produce ethyne



(b) From Vicinal Dihalides

Vicinal dihalides first treatment with aq. KOH & then reaction with NaNH_2 produces alkyne



Physical Properties of Alkynes

1. First 3 are gases, next 8 liquids, higher ones solids.
2. Colourless and odourless (except acetylene – garlic smell).
3. Melting/boiling points and densities slightly higher than alkenes/alkanes; increase with molecular mass.
4. Lighter than water, insoluble in it but soluble in non-polar solvents; weakly polar in nature.

CHEMICAL RXNS OF ALKYNE.

Acidic
character

Addition Rxn.

- H_2
- X_2
- HX
- H_2O

Polymerisation

Chemical Properties of Alkynes



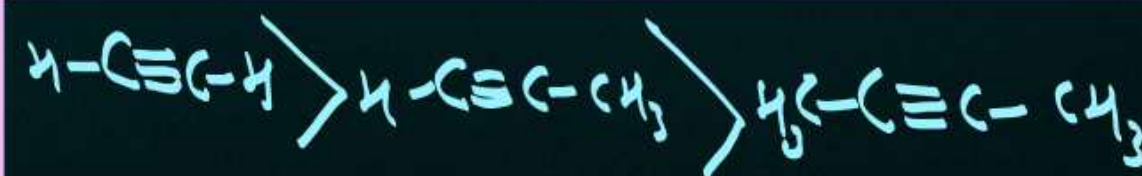
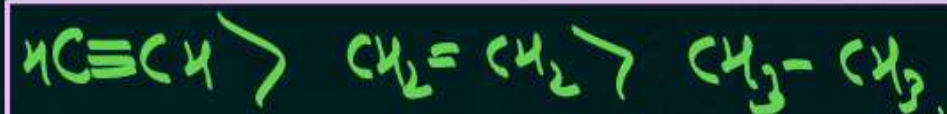
(a) Acidic Character of Alkyne



- In ethyne (and other alkynes), the carbon atom is sp -hybridized, which has the highest s -character and therefore the highest electronegativity. This pulls the $C-H$ bond electrons closer to carbon, making it easier to remove the hydrogen as a proton (H^+). That's why the hydrogens attached to a triple-bonded carbon are acidic in nature.



ORDER OF ACIDIC CHARACTER.



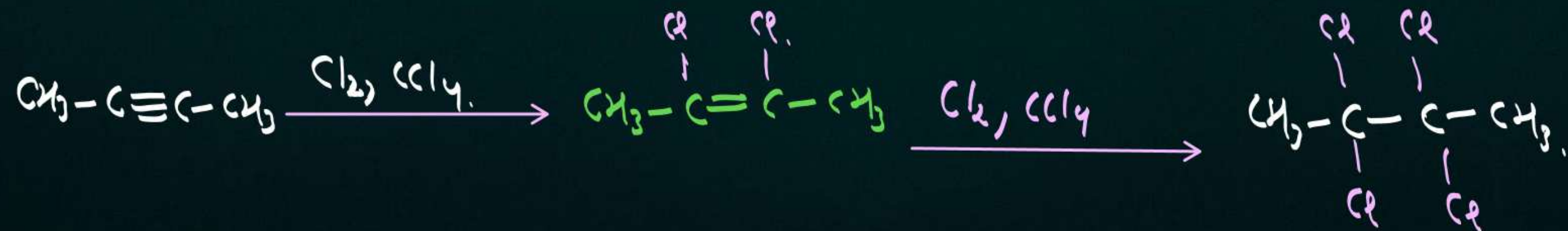
It undergoes addition rxn just like alkene.

(1) Addition of Dihydrogen

$$\text{H}-\text{C}\equiv\text{C}-\text{H} \xrightarrow{\text{H}_2, \text{Pd, Ni}} \text{H}-\text{C}=\text{C}-\text{H} \xrightarrow{\text{H}_2, \text{Pd, Ni}} \text{H}-\text{C}-\text{C}-\text{H}$$

Ethyne Ethene Ethane

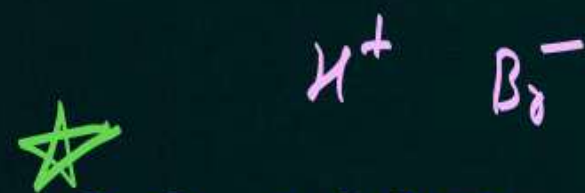
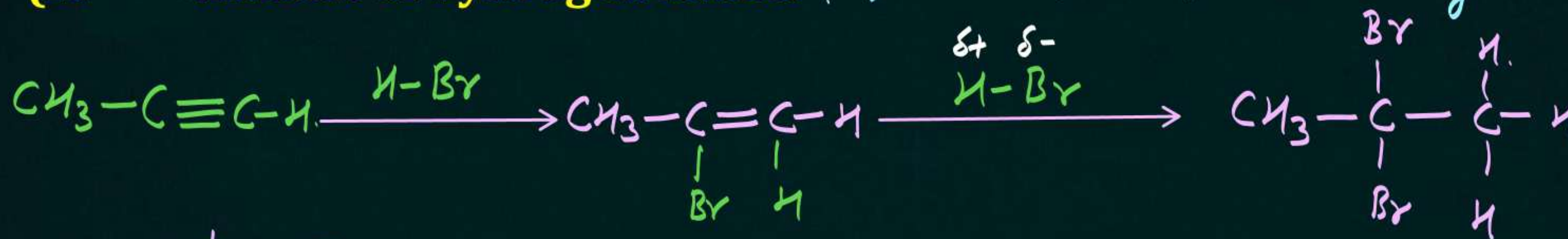
(ii) Addition of Halogens $\rightarrow$ But an excess of halogen tetrahalides are formed.



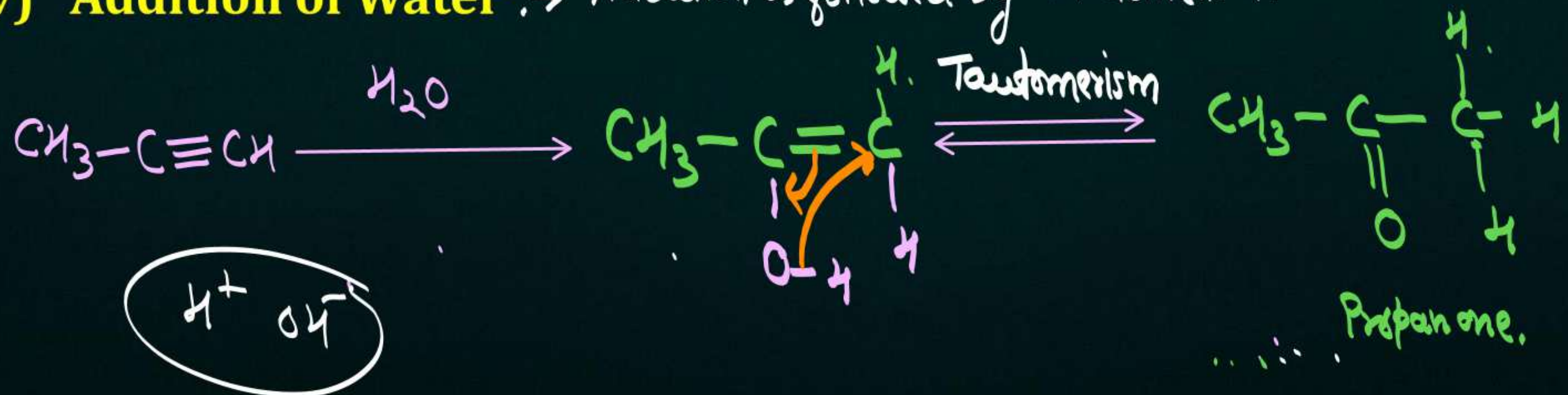
Chemical Properties of Alkynes

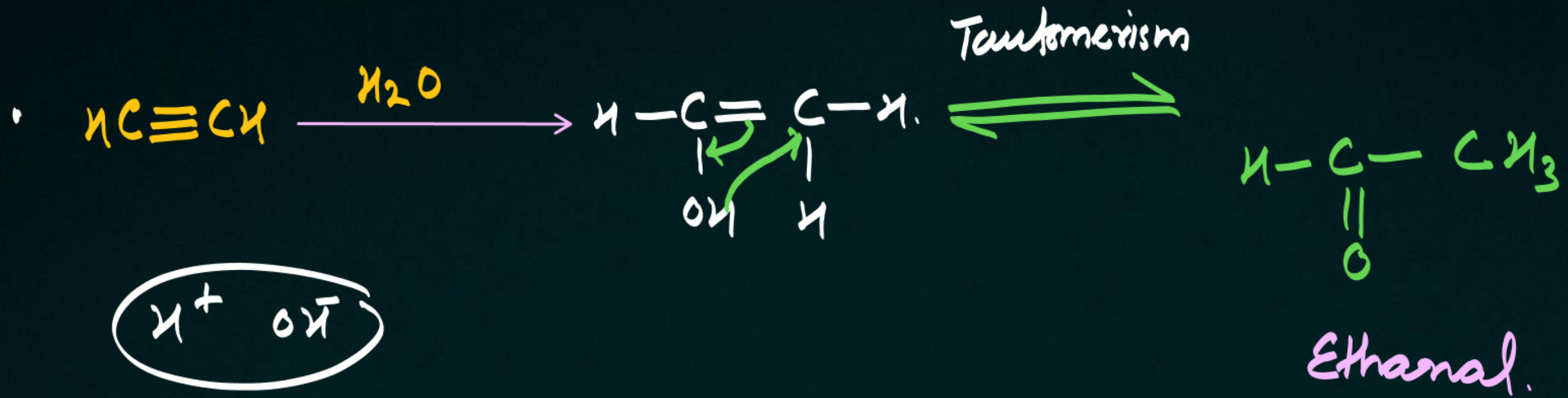


(iii) **Addition of hydrogen halide** $\rightarrow$ Addition takes place according to Markovnikov's rule



(iv) **Addition of Water** $\rightarrow$ Addition is followed by Tautomerism.





QUESTION



The geometry of ethyne molecule is:

- 1 Linear
- 2 Trigonal planar
- 3 Tetrahedral
- 4 Bent

QUESTION

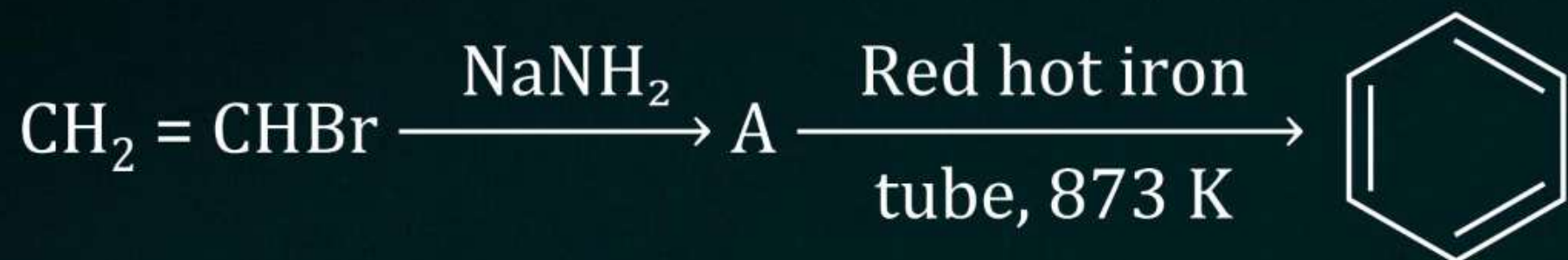
Consider the following reaction,



Here X refers to

- 1 Ethene
- 2 Ethyne
- 3 Methyne
- 4 None of these

QUESTION



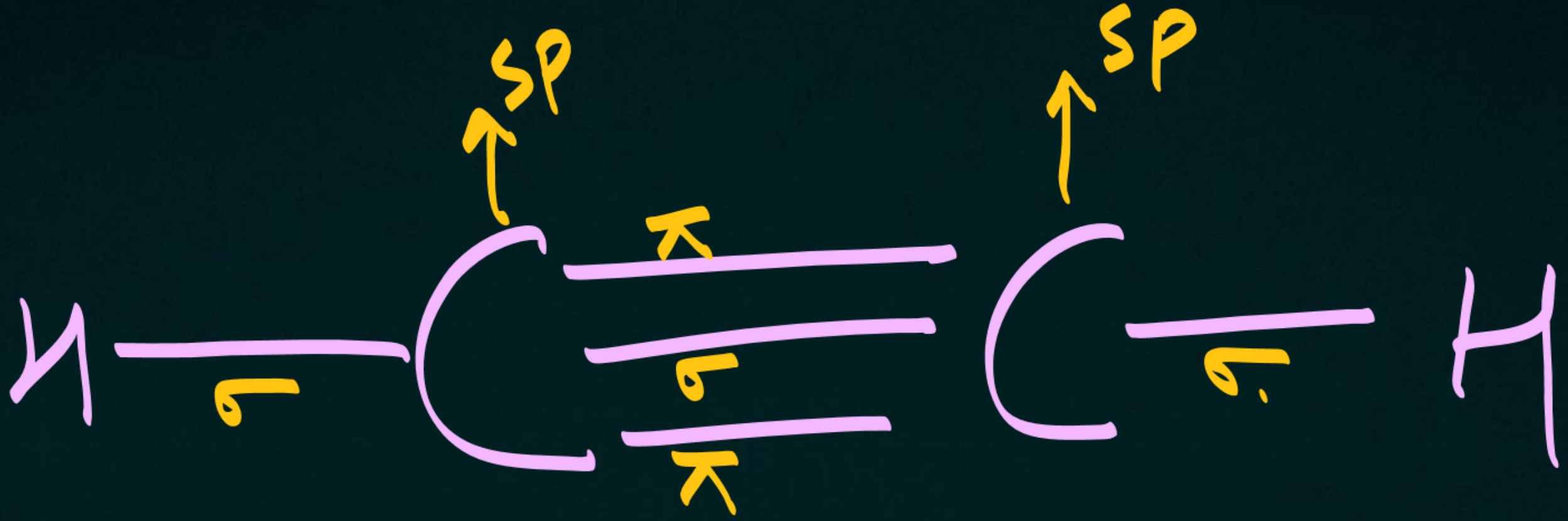
Here, A = refers to

- 1 $\text{CH} \equiv \text{CH}$
- 2 $\text{CH}_3 - \text{C} \equiv \text{CH}$
- 3 $\text{CH}_3 - \text{CH}_3$
- 4 $\text{CH} \equiv \text{C} - \text{NH}_2$

QUESTION

When $\text{CH}_3 - \text{C} \equiv \text{CH}$ reacts with **one mole of HBr** then product obtained is

- 1 Bromoethene
- 2 Dibromopropane
- 3 2-Bromopropene
- 4 1-Bromopropane



- 1) Numaan Wani
- 2) UZAIR SONAIL
- 3) Usman YUSUF.

Thank
You